

A Near-Logarithmic Quantitative Ornstein Lower Bound

Automated Math Discovery Run `fa.banach.001`, lane 4

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Abstract

Let K_N be the least constant for which every trigonometric polynomial of coordinate degree at most N on $\mathbb{T}^2$ satisfies

$$\|\partial_x \partial_y f\|_1 \leq K_N (\|\partial_x^2 f\|_1 + \|\partial_y^2 f\|_1).$$

Kazaniecki and Wojciechowski proved $K_N \gtrsim \sqrt{\log N}$, recalled the upper bound $K_N \lesssim \log N$, and conjectured sharp logarithmic growth. We prove the near-sharp improvement

$$K_N \gtrsim \frac{\log N}{\log \log N}.$$

The gain comes from estimating the rational Fourier multipliers blockwise on a geometrically lacunary Riesz product rather than summing the errors of its individual Fourier coefficients.

1 Source question and result

The source's simplest quantitative Ornstein problem concerns the mixed second derivative versus the two pure second derivatives. It proves the lower exponent $1/2$, gives a weak-type argument for the logarithmic upper bound, and asks whether the latter is sharp; see Figure 1.

For definiteness, the degree of a trigonometric polynomial on $\mathbb{T}^2$ is the maximum absolute value of either coordinate in its Fourier support. Define

$$K_N = \sup_{\substack{f \neq 0 \\ \deg f \leq N}} \frac{\|\partial_x \partial_y f\|_{L^1(\mathbb{T}^2)}}{\|\partial_x^2 f\|_{L^1(\mathbb{T}^2)} + \|\partial_y^2 f\|_{L^1(\mathbb{T}^2)}}.$$

Theorem 1. *There are universal constants $c > 0$ and N_0 such that, for $N \geq N_0$,*

$$K_N \geq c \frac{\log N}{\log \log N}.$$

This does not prove the conjectured $K_N \asymp \log N$, but it leaves only one iterated logarithm between the new lower bound and the known upper bound.

2 Idea of the proof

Take lacunary frequencies

$$a_k = (M^k, (-1)^k M^k), \quad 1 \leq k \leq n,$$

Corollary 1. *For every $N \in \mathbb{N}$ there exists a trigonometric polynomial P_N on $\mathbb{T}^2$ of degree N , which satisfies*

$$\left\| \frac{\partial^2}{\partial x^2} P_N \right\|_{L_1(\mathbb{T}^d)} + \left\| \frac{\partial^2}{\partial y^2} P_N \right\|_{L_1(\mathbb{T}^d)} \leq 1$$

but

$$\left\| \frac{\partial^2}{\partial x \partial y} P_N \right\|_{L_1(\mathbb{T}^d)} > C \ln^{\frac{1}{2}} N.$$

We do not know if the bound from Corollary 1 is sharp. In fact one can establish in a rather standard way that the mixed derivative from Corollary 1 could not have norm greater than $c \ln n$. Indeed the (linear and invariant) operator T which retrieves the mixed derivative from the pure ones is of a weak type $(1, 1)$ (see [5]). Hence, by the Nikolskii type inequality for Lorentz spaces (see [15] Theorem 3] and [1] Lemma 3.1]), for a trigonometric polynomial f of degree N

$$\begin{aligned} \left\| \frac{\partial^2}{\partial x \partial y} f \right\|_{L_1(\mathbb{T}^2)} &= \|Tf\|_{L_{1,1}(\mathbb{T}^2)} \leq C \ln(1+N) \|Tf\|_{L_{1,\infty}(\mathbb{T}^2)} \\ &\leq C \ln(1+N) \left(\left\| \frac{\partial^2}{\partial x^2} f \right\|_{L_1(\mathbb{T}^2)} + \left\| \frac{\partial^2}{\partial y^2} f \right\|_{L_1(\mathbb{T}^2)} \right). \end{aligned}$$

The same comment concerns all the results obtained in this paper. All bounds from below presented here are of the form $(\ln N)^\phi$ for some $\phi < 1$, while the common upper bound is $\ln N$. Nevertheless we conjecture that the optimal exponent ϕ should be equal to one (see Remark 8).

This paper contains final results of the study we began in [8]. It seems it is the first use in the literature of trigonometric polynomials in the context of Ornstein non-inequalities.

Figure 1: The sharpness question in the source paper, arXiv:2206.13666.

and the Riesz product $P_n = \prod_{k=1}^n (1 + \cos \langle a_k, z \rangle)$. Write $R_n = P_n - 1$ as the sum of its martingale-difference blocks B_k . We construct W_n by inverting ∂_x^2 on the Fourier support of R_n . On the k th block, the multipliers of the other two derivatives are

$$\frac{q_y}{q_x} \approx (-1)^k, \quad \left(\frac{q_y}{q_x} \right)^2 \approx 1.$$

Thus $\partial_y^2 W_n$ is close to R_n , whose L^1 norm is at most two, whereas $\partial_x \partial_y W_n$ is close to an alternating transform of the Riesz blocks, whose L^1 norm is comparable to n .

The point is the error estimate. Earlier frequencies contribute a relative amount $O(1/M)$ to a block frequency. Expanding the rational multipliers as power series and using the L^1 Bernstein inequality bounds a whole block's error by $O(1/M)$, without paying for its exponentially many Fourier coefficients. Taking $M = n$ makes the total error $O(1)$. The degree is then at most $2n^n$, so n is of order $\log N / \log \log N$.

3 The block multiplier estimate

Put $\epsilon_k = (-1)^k$, $a_k = (M^k, \epsilon_k M^k)$, and

$$P_0 = 1, \quad P_k = \prod_{j=1}^k (1 + \cos \langle a_j, z \rangle), \quad B_k = P_k - P_{k-1}.$$

Then

$$B_k = \cos \langle a_k, z \rangle P_{k-1}, \quad R_n = P_n - 1 = \sum_{k=1}^n B_k.$$

Since $M \geq 3$, the first coordinate of every nonzero frequency in these blocks is nonzero. On their union define Fourier multipliers

$$\widehat{T_1 h}(q) = \frac{q_y}{q_x} \widehat{h}(q), \quad \widehat{T_2 h}(q) = \left(\frac{q_y}{q_x} \right)^2 \widehat{h}(q).$$

Lemma 2 (Blockwise error). *There is a universal C such that, whenever $M \geq 4$,*

$$\left\| T_1 R_n - \sum_{k=1}^n \epsilon_k B_k \right\|_1 \leq C \frac{n}{M-1}, \quad \|T_2 R_n - R_n\|_1 \leq C \frac{n}{M-1}.$$

Proof. Fix k and one of the two signed halves of B_k . Its frequencies have the form

$$q = \sigma a_k + s, \quad \sigma \in \{-1, 1\}, \quad s \in \text{supp } \widehat{P}_{k-1}.$$

Set

$$u = \frac{s_x}{\sigma M^k}, \quad v = \frac{s_y}{\sigma M^k}, \quad \delta_k = \frac{\sum_{j < k} M^j}{M^k} < \frac{1}{M-1}.$$

On this half-block,

$$\frac{q_y}{q_x} = \frac{\epsilon_k + v}{1 + u}.$$

We interpret a monomial $u^r v^s$ as a normalized differential multiplier on P_{k-1} . The one-variable Bernstein inequality in L^1 , applied successively in x and y , gives

$$\|u^r v^s P_{k-1}\|_1 \leq \delta_k^{r+s} \|P_{k-1}\|_1 = \delta_k^{r+s}.$$

Here $P_{k-1} \geq 0$ and its integral is one; lacunarity ensures that its constant Fourier coefficient is one.

Now expand

$$\frac{\epsilon_k + v}{1 + u} - \epsilon_k = \frac{v - \epsilon_k u}{1 + u} = \sum_{r \geq 0} (-u)^r (v - \epsilon_k u).$$

The preceding Bernstein estimate and absolute summation show that the T_1 error on the entire k th block is at most

$$\frac{2\delta_k}{1 - \delta_k}.$$

For the square multiplier, use

$$\left(\frac{\epsilon_k + v}{1 + u} \right)^2 - 1 = \frac{2\epsilon_k v + v^2 - 2u - u^2}{(1 + u)^2}, \quad (1 + u)^{-2} = \sum_{r \geq 0} (-1)^r (r + 1) u^r.$$

Its block error is at most

$$\frac{4\delta_k + 2\delta_k^2}{(1 - \delta_k)^2}.$$

The factors $1/2$ from the two signed halves add to one. Summing these uniform $O((M-1)^{-1})$ bounds over k proves both assertions. $\square$

4 The alternating Riesz transform

We use the same two standard ingredients invoked in the source paper. To make the transference precise, write a polynomial on the Riesz spectrum as

$$F(x, y) = \sum_{\xi \in \{-1, 0, 1\}^n} c_\xi \exp\left(i \sum_{k=1}^n \xi_k (M^k x + \epsilon_k M^k y)\right).$$

For each fixed y , Meyer's lacunary transference theorem applied to the x -frequencies M^k compares $\|F(\cdot, y)\|_{L_x^1}$ with

$$\left\| \sum_{\xi} c_\xi \exp\left(i \sum_{k=1}^n \xi_k (t_k + \epsilon_k M^k y)\right) \right\|_{L^1(\mathbb{T}_t^n)}.$$

The latter is independent of y , by the Haar translations $t_k \mapsto t_k - \epsilon_k M^k y$. Integrating in y therefore compares the $L^1(\mathbb{T}^2)$ norm with the fully independent model. The comparison constant depends only on the sum of consecutive frequency ratios, and here

$$\sum_{k < n} \frac{M^k}{M^{k+1}} = \frac{n-1}{M}.$$

Hence the comparison constant is universal when $M \geq n$.

In the independent model, the blocks are

$$\tilde{B}_k = \tilde{P}_k - \tilde{P}_{k-1}, \quad \tilde{P}_k = \prod_{j=1}^k (1 + \cos t_j).$$

Summation by parts gives

$$\sum_{k=1}^n \epsilon_k \tilde{B}_k = \epsilon_n \tilde{P}_n - \epsilon_1 \tilde{P}_0 + \sum_{k=1}^{n-1} (\epsilon_k - \epsilon_{k+1}) \tilde{P}_k.$$

The lower inequality for linear combinations of finite Riesz products due to Latała (in the form used by Kazaniecki–Wojciechowski) controls the L^1 norm from below by a universal constant times the total variation of these coefficients. Since $|\epsilon_k - \epsilon_{k+1}| = 2$, transference yields:

Lemma 3 (Alternating main term). *There is a universal $c_0 > 0$ such that, for $M \geq n \geq 2$,*

$$\left\| \sum_{k=1}^n \epsilon_k B_k \right\|_1 \geq c_0 n.$$

5 Proof of Theorem 1

Fix $n \geq 4$ and set $M = n$. Define a real trigonometric polynomial W_n by

$$\widehat{W}_n(q) = -\frac{\widehat{R}_n(q)}{q_x^2} \quad (q \in \text{supp } \widehat{R}_n), \quad \widehat{W}_n(q) = 0 \quad \text{otherwise.}$$

The domination of the last lacunary digit gives $|q_x| \geq M^k - \sum_{j < k} M^j > 0$ on the k th block, so this is well defined. Fourier differentiation gives the exact identities

$$\partial_x^2 W_n = R_n, \quad \partial_x \partial_y W_n = T_1 R_n, \quad \partial_y^2 W_n = T_2 R_n.$$

Because $P_n \geq 0$ and $\int P_n = 1$,

$$\|R_n\|_1 = \|P_n - 1\|_1 \leq 2.$$

Lemmas 2 and 3, with $M = n$, now imply

$$\|\partial_y^2 W_n\|_1 \leq C_1, \quad \|\partial_x \partial_y W_n\|_1 \geq c_0 n - C_2.$$

After increasing the lower threshold for n ,

$$\frac{\|\partial_x \partial_y W_n\|_1}{\|\partial_x^2 W_n\|_1 + \|\partial_y^2 W_n\|_1} \geq c_1 n.$$

Every Fourier coordinate of R_n , and hence of W_n , has magnitude at most

$$\sum_{k=1}^n M^k \leq 2M^n = 2n^n.$$

Thus $K_{2n^n} \geq c_1 n$. Finally, for sufficiently large N , take

$$n = \left\lfloor \frac{\log N}{2 \log \log N} \right\rfloor.$$

Then $2n^n \leq N$. Since K_N is nondecreasing,

$$K_N \geq c_1 n \geq c \frac{\log N}{\log \log N},$$

which proves the theorem. □

6 Scope and remaining gap

The proof is specific to the basic isotropic pair $(\partial_x^2, \partial_y^2)$ and target $\partial_x \partial_y$. The same blockwise principle may improve other configurations in the source paper, but no such generalization is claimed here. Achieving the conjectured full $\log N$ lower bound would require uniformly controlling the pure-derivative error at fixed geometric lacunarity; the present absolute block summation costs n/M and forces $M \asymp n$, exactly producing the $\log \log N$ loss.

References

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